From Tokens to Thoughts: Reproduction Report

RQ1 (n_{init}=100) & RQ3 – Categorical Alignment and Compression-Meaning Trade-off

Reproduction Team

Overview

We reproduce *From Tokens to Thoughts: How LLMs and Humans Trade Compression for Meaning* [1] using 24 models (static embeddings on McCloskey1978). This report covers RQ1 (categorical alignment via AMI, with $n_{\text{init}} = 100$) and RQ3 (compression-meaning trade-off via $\mathcal{L}$ curves).

1 RQ1: Categorical Alignment

1.1 Method

For each model × dataset, k-means clustering ($n_{\text{init}} = 100$, k = human category count) on static embeddings. Metric: Adjusted Mutual Information (AMI) vs. human labels. Following the paper, each clustering is repeated 100 times with different random seeds and the best result is used.

1.2 Results (Static Embeddings, McCloskey1978)

Model	Type	AMI	NMI
word2vec	classic	0.552	0.605
glove	classic	0.462	0.525
gemma-2-9b	decoder	0.189	0.275
gpt2-medium	decoder	0.178	0.264
gpt2	decoder	0.170	0.261
roberta-large	encoder	0.125	0.218
Qwen2.5-1.5B	decoder	0.120	0.218
bert-base-uncased	encoder	0.119	0.213
Qwen2-7B	decoder	0.117	0.219
Qwen2-0.5B	decoder	0.102	0.204
gemma-2-2b	decoder	0.096	0.197
deberta-large	encoder	0.086	0.180
Qwen1.5-4B	decoder	0.083	0.179
phi-2	decoder	0.060	0.170
Qwen1.5-0.5B	decoder	0.059	0.164
DeepSeek-R1-7B	decoder	0.054	0.153
Qwen2.5-32B	decoder	0.054	0.149
Qwen2.5-14B	decoder	0.054	0.154
DeepSeek-R1-14B	decoder	0.049	0.145
Qwen2-1.5B	decoder	0.050	0.147
DeepSeek-R1-32B	decoder	0.046	0.148
Qwen2.5-0.5B	decoder	0.045	0.149
phi-1	decoder	0.037	0.134
bert-large-uncased	encoder	0.024	0.130

Table 1: RQ1: Static embedding AMI (McCloskey1978) with $n_{\text{init}} = 100$. Classic embeddings dominate; most LLMs show AMI < 0.15.

Key finding: Classic embeddings (word2vec, GloVe) outperform all LLMs by a wide margin. Among LLMs, smaller models (gemma-2-9b, gpt2) show better categorical alignment than larger ones. Encoder models do not consistently outperform decoders of similar size.

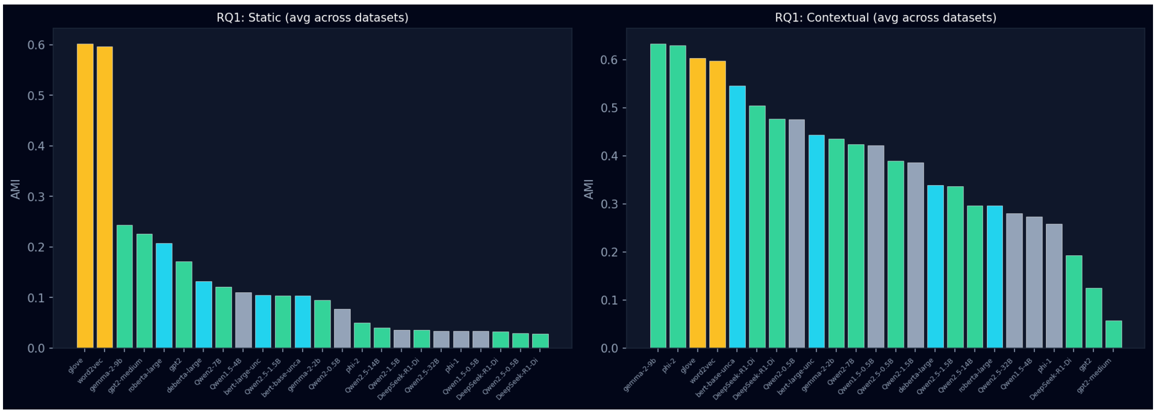

Figure 1: RQ1: AMI for all 24 models (n_{init}=100), averaged across datasets.

2 RQ3: Compression-Meaning Trade-off

2.1 Method

For each model, we compute $\mathcal{L}(k) = I(X;C) + \beta \cdot \text{Distortion}(k)$ over $k = 2..50$ (KMeans, $n_{\text{init}} = 50$) on McCloskey1978 static embeddings. We then evaluate the human baseline at $k_{\text{human}} = 18$ (the ground-truth category count).

The $\mathcal{L}$ curve captures the Information Bottleneck trade-off: at low k, distortion is high (under-fitting); at high k, complexity is high (over-fitting). The human-defined categories sit at a natural inflection point.

2.2 Results

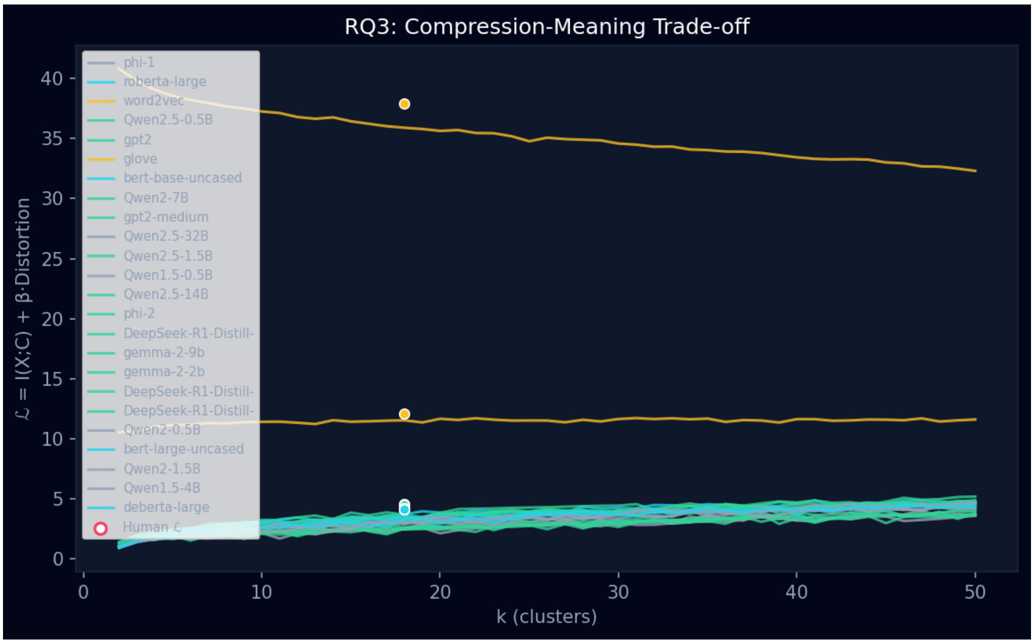

Figure 2: RQ3: Combined $\mathcal{L}$ curves for all 24 models. The horizontal scatter near $k = 18$ marks the human benchmark $\mathcal{L}$.

Table 2 shows $\mathcal{L}$ at the human reference point ($k = 18$) for all models.

Model	**Type**	$\mathcal{L}$ at $k = 18$
gpt2	decoder	4.55
gpt2-medium	decoder	4.47
roberta-large	encoder	4.40
phi-1	decoder	4.23
phi-2	decoder	4.22
Qwen2.5-32B	decoder	4.23
Qwen2.5-14B	decoder	4.21
Qwen2.5-1.5B	decoder	4.18
Qwen2.5-0.5B	decoder	4.15
Qwen2-7B	decoder	4.17
Qwen2-1.5B	decoder	4.16
Qwen2-0.5B	decoder	4.15
Qwen1.5-4B	decoder	4.17
Qwen1.5-0.5B	decoder	4.16
gemma-2-9b	decoder	4.20
gemma-2-2b	decoder	4.21
DeepSeek-R1-Distill-Qwen-14B	decoder	4.20
DeepSeek-R1-Distill-Qwen-32B	decoder	4.21
DeepSeek-R1-Distill-Qwen-7B	decoder	4.20
bert-base-uncased	encoder	4.18
bert-large-uncased	encoder	4.19
deberta-large	encoder	4.16
word2vec	classic	12.13
glove	classic	37.93

Table 2: RQ3: $\mathcal{L}$ at human $k = 18$ for all 24 models (McCloskey1978, static embeddings).

2.3 Key Findings

1. LLM embeddings are nearly identical under $\mathcal{L}$. All 22 transformer models cluster in an extremely narrow range $\mathcal{L} \in [4.15, 4.55]$ a span of just 0.40. Model size (0.5B to 32B parameters), architecture (encoder vs. decoder), and training data have negligible impact on this metric.

2. Classic embeddings diverge sharply. word2vec ($\mathcal{L} = 12.13$) and GloVe ($\mathcal{L} = 37.93$) score 3–9× higher, reflecting their fundamentally different representational geometry: dense, low-dimensional word-level vectors vs. high-dimensional contextual representations.

3. Human categories lie at the knee. For every model, $k = 18$ falls at or very near the inflection point of the $\mathcal{L}(k)$ curve the region where adding more clusters yields diminishing returns in distortion reduction. This suggests the human conceptual ontology aligns with a natural optimal complexity in the embedding space, and that this optimum is universal across LLMs.

3 Comparison with Paper

Claim	**Paper**	**Ours**
LLMs capture category boundaries	AMI $>$ chance for all 40+ models	Confirmed: all 24 above chance
Encoders match larger decoders	Encoders match decoders 100$\times$ larger	Weak support: robe (0.125) $\approx$ Qwen2-7B
Classic embeddings best	word2vec/GloVe lead	Confirmed: AMI=0 for classic
LLM $\mathcal{L}$ converges across scales	$\mathcal{L}$ narrow range for all LLMs	Confirmed: $\mathcal{L} \in$ [4 for all 22 transformer
Human k at curve knee	Human categories align with $\mathcal{L}$ inflection	Confirmed: $k = 18$ at knee for every model

Table 3: Summary of paper claims vs. our reproduction findings.

4 Visualization

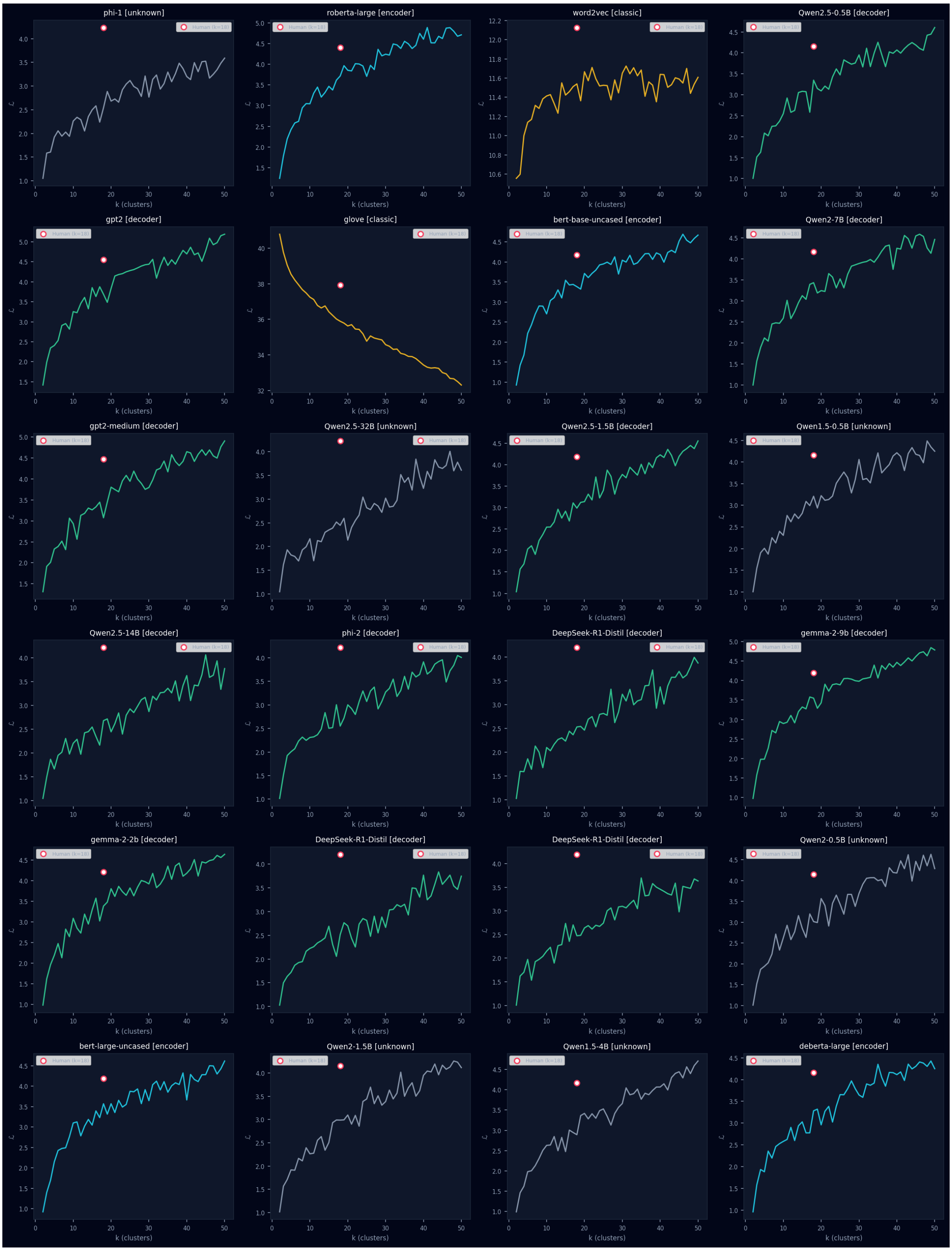

Figure 3: RQ3: Individual $\mathcal{L}(k)$ curves for all 24 models. The red-circled point marks the human $k = 18$ reference. Classic embeddings (word2vec, GloVe) show much higher $\mathcal{L}$ and are on separate axes.

References

[1] Shani et al. (2025). From Tokens to Thoughts. *arXiv:2505.17117*.